

A second-tier H^2 obstruction for supply-chain inventory: the inventory gerbe

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Abstract

A supply chain, abstracted to a small category $\mathcal{C}$, carries inventory as local stock-data glued along logistics operations. When charts are compared by *equality* of counts, we showed earlier that consistency is a degree-one class $[\theta_R] \in H^1(N(\mathcal{C}); R)$. Here we refine the comparison to *isomorphism of stock-states*: lots are fungible, so charts match only up to a relabelling group $K = \text{Aut}(F)$. We prove that this refinement turns the gluing data into a K -gerbe over the nerve $N(\mathcal{C})$ and raises the obstruction by exactly one categorical level. A strict global inventory exists if and only if a single degree-two class $[\mathbf{g}] \in H^2(N(\mathcal{C}); Z(\text{Aut}F))$ vanishes (Theorem A); $[\mathbf{g}]$ is genuinely degree-two — it vanishes on every acyclic (DAG or tree) chain yet is already nonzero on a single reversible two-lot loop $B(\mathbb{Z}/2)$ (Theorem B); and it is logically independent of the Zappa–Szép weld class $[\omega]$, which lives in the *same* degree but over a *different* coefficient system (Theorem C). Finally, $[\mathbf{g}]$ can be nonzero only when $Z(\text{Aut}F) \neq 1$, so generic ≥ 3 -fold fungibility carries no tier-two obstruction while gauge-structured stock does (Proposition D). The result completes inventory consistency into a two-tier theory — H^1 counts, H^2 relabellings — and exhibits a fourth distinct H^2 home on $\text{DCont} \simeq \text{Cat}$.

1 Introduction

Composition is where correctness is won or lost, and inventory is where the price is paid. When two delivery routes reconverge on the same stock-keeping unit, when stock is moved and merged and counted again downstream, the question is always whether the local records glue to one consistent global record. When they do not, the failure has an economic name: phantom inventory, a reconciliation that never closes, a write-off. A compositional theory of supply chains earns its keep by telling you *which* such failures can occur and *where* they live.

The categorical model is standard. A supply chain — nodes as objects, admissible logistics operations as morphisms — is a small category $\mathcal{C}$; equivalently, by Ahman and Uustalu [1], a *directed container*. Its nerve $N(\mathcal{C})$ is the simplicial set whose n -simplices are composable strings of operations, and the cohomology of $N(\mathcal{C})$ is the natural home for obstructions to gluing local data along $\mathcal{C}$.

In a companion note [2] we asked when inventory, recorded as counts transported along operations, glues consistently. Writing the inventory as a resource presheaf R on $\mathcal{C}$, the answer was clean:

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consistency is the vanishing of a *degree-one* class $[\theta_R] \in H^1(N(\mathcal{C}); R)$, the route-reconvergence discrepancy of stock counts, and this class is logically independent of the Zappa–Szép weld obstruction $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ that governs whether $\mathcal{C}$ factors as a product of two chains at all [3, 4]. That note flagged one refinement as the natural next question, and the present note answers it.

The gap. Equality of counts is the wrong comparison when lots are *fungible*. Two pallets of the same grade are interchangeable; a local stock-state is a labelled configuration, but the labels carry no information, so two charts should be regarded as matching when they are *isomorphic*, not when they are literally equal. Each inventory fibre F then carries a *relabelling group* $K := \mathrm{Aut}(F)$, and comparison of local charts is canonical only up to K . Gluing up-to-isomorphism is not a gluing-of-sets problem; it is a gluing-of-*stacks* problem — a gerbe. The companion note anticipated that the obstruction would then rise to H^2 , but neither the coefficient system nor its relationship to the weld was determined. Does the obstruction equal $[\omega]$ (which also lives in H^2)? What exactly must vanish for a strict global inventory to exist? For which fibres is the phenomenon even nonvacuous?

The results. We make the refinement precise and settle all three questions. Fix a fibre F with relabelling group $K = \mathrm{Aut}(F)$, centre $Z := Z(K)$, and inner group $\mathrm{Inn}K := K/Z$, giving the central extension

$$1 \longrightarrow Z \longrightarrow K \xrightarrow{\pi} \mathrm{Inn}K \longrightarrow 1. \quad (\star)$$

Under a scope hypothesis stated below (the gerbe’s band is trivialised), the gluing data is a K -gerbe over $N(\mathcal{C})$ presented by a normalised 2-cochain $\omega \in C^2(N(\mathcal{C}); Z)$, and its class $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z)$ is the tier-two obstruction. Our main results are:

Theorem 1 (Rectification $\Leftrightarrow$ vanishing; Theorem A). *A strict global inventory — a strict functor $\mathcal{C} \rightarrow BK$ realising the up-to-iso gluing — exists if and only if $[\mathfrak{g}] = 0$ in $H^2(N(\mathcal{C}); Z)$.*

Theorem 2 (Degree-genuineness; Theorem B). *If $N(\mathcal{C})$ is homotopy 1-truncated — in particular if $\mathcal{C}$ is the free category on a directed acyclic quiver (a tree-like or DAG chain) — then $H^2(N(\mathcal{C}); Z) = 0$, so $[\mathfrak{g}] = 0$ for every fibre. There is a minimal one-node witness, $\mathcal{C} = B(\mathbb{Z}/2)$ with $K = \mathbb{Z}/2$, for which $[\mathfrak{g}] \neq 0$.*

Theorem 3 (Independence from the weld; Theorem C). *$[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(\mathrm{Aut}F))$ and the Zappa–Szép weld $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ are logically independent: neither vanishing implies the other. They are classes of the same degree over different coefficient systems — the fibre centre $Z(\mathrm{Aut}F)$ versus the weld vertex-group presheaf $\mathcal{D}$ — and we exhibit chains realising each of $([\mathfrak{g}] \neq 0, [\omega] = 0)$ and $([\mathfrak{g}] = 0, [\omega] \neq 0)$.*

Proposition 4 (Sharp nonvacuousness; Proposition D). *$[\mathfrak{g}]$ can be nonzero only if $Z(\mathrm{Aut}F) \neq 1$. For F a bare n -element set of indistinguishable lots, $\mathrm{Aut}(F) = S_n$ has $Z(S_n) = 1$ for $n \geq 3$; hence generic ≥ 3 -fold fungibility carries no tier-two obstruction, and the gerbe is a phenomenon of two-fold or gauge/torsor-structured ambiguity.*

The main result is thus visible by page two, and the picture is a *two-tier* obstruction. Tier one compares stock by equality; tier two compares it up to relabelling; the two live in consecutive degrees over different coefficients, and the weld is a third, structural precondition orthogonal to both:

tier	charts compared by	obstruction	coefficients, degree
1	equality of counts	$[\theta_R]$	R (resource group), H^1
2	iso of stock-states (relabelling)	$[\mathfrak{g}]$	$Z(\text{Aut}F)$ (fibre centre), H^2
(precond.)	does $\mathcal{C}$ factor at all	$[\omega]$	$\mathcal{D}$ (weld vertex groups), H^2

The method. Two standard facts, specialised to $N(\mathcal{C})$, do the work. First, the *band* of a band-trivialised K -gerbe is the centre $Z(K)$ with trivial $\mathcal{C}$ -action (Lemma 7): inner automorphisms can be undone by re-choosing local trivialisations, so only central ambiguity survives global comparison. Second, when K is nonabelian the unambiguous datum is a functor $\rho : \mathcal{C} \rightarrow B(\text{Inn}K)$, and the gerbe is its *gerbe of lifts* through $(\star)$; choosing set-section lifts $T_f = s(\rho_f)$ exhibits a Z -valued 2-cocycle ω whose class is $[\mathfrak{g}]$. Rectification then reads as trivialising ω by a central 1-cochain — an elementary coboundary computation (Theorem A) that is uniform across the abelian and nonabelian faces. The remaining theorems are a simplicial computation on the nerve (Theorem B), a coefficient comparison with explicit witnesses (Theorem C), and the classical centre table (Proposition D).

Consequences and placement. Three points are worth stating up front. (i) The refinement is *exactly* one categorical level: equality-gluing is H^1 , iso-gluing is H^2 , and nothing in between. (ii) Because $[\mathfrak{g}]$ and $[\omega]$ share a degree, the temptation to identify “inventory is inconsistent up to relabelling” with “the chain fails to weld” is now a *degree-matched* trap, and the only discriminator is the coefficient system; Theorem C checks it. (iii) The nonvacuousness criterion sharpens the applications story: the economically relevant tier-two case is not plain fungibility of many interchangeable units (centreless S_n) but *gauge-valued* stock — allocation defined up to a group action — whose automorphism group has a centre. Within the Kodamai program, $[\mathfrak{g}]$ is the *fourth* distinct H^2 home on $\text{DCont} \simeq \text{Cat}$, after the Zappa–Szép holonomy, Ferri prunability, and the Baues–Wirsching class H^2_{Cat} ; unlike the first three it is distinguished by *coefficients*, not by site.

Outline. Section 2 recalls supply chains as directed containers, the central extension $(\star)$, and the gerbe vocabulary we use. Section 3 sets up the tier-two datum, proves the band is the centre (Lemma 7), and exhibits the cocycle in both the abelian and nonabelian faces. Section 4 proves Theorem A. Section 5 proves degree-genuineness (Theorem B). Section 6 proves independence from the weld (Theorem C). Section 7 proves the nonvacuousness criterion (Proposition D). Section 8 states the scope, the honest open problem (a nontrivial outer band), and the two-tier bottom line.

2 Preliminaries

The section records only what the proofs use. A reader fluent in directed containers and nonabelian H^2 can skip to Section 3.

2.1 Supply chains as directed containers and their nerve

A supply chain is modelled as a small category $\mathcal{C}$: objects are nodes (warehouses, plants, stock locations), morphisms are admissible logistics operations, composition is sequencing. By Ahman–Uustalu [1] a small category is the same thing as a directed container, so the model lives natively in the container program.

The *nerve* $N(\mathcal{C})$ is the simplicial set with $N(\mathcal{C})_0 = \text{Ob}\mathcal{C}$, $N(\mathcal{C})_1 = \text{Hom}\mathcal{C}$, and $N(\mathcal{C})_n$ the set of composable strings $c_0 \xrightarrow{f_1} c_1 \rightarrow \dots \xrightarrow{f_n} c_n$. For an abelian coefficient group A with (here) trivial

$\mathcal{C}$ -action, $H^\bullet(N(\mathcal{C}); A)$ is the cohomology of the normalised cochain complex $C^n(N(\mathcal{C}); A) = \{ u : N(\mathcal{C})_n \rightarrow A \mid u \text{ vanishes on degeneracies} \}$ with the usual alternating-face differential. Two facts are used below:

- For a one-object category $\mathcal{C} = BG$ (a group G viewed as a category), $N(BG)$ is the standard classifying space and $H^\bullet(N(BG); A) = H^\bullet(G; A)$ is group cohomology.
- For the free category $F(Q)$ on a quiver Q , the classifying space $|N(F(Q))|$ is homotopy equivalent to the geometric realisation $|Q|$, a 1-complex; hence $H^{\geq 2}(N(F(Q)); A) = 0$.

2.2 Inventory fibres and the central extension

An *inventory fibre* F is the local type of a stock-state at a node: a configuration of lots. Fungibility means the labels carry no information, so the relevant symmetry is the *relabelling group*

$$K := \text{Aut}(F), \quad Z := Z(K) \text{ (centre)}, \quad \text{Inn}K := K/Z \text{ (inner group)},$$

fitting into the central extension $(\star)$. We write $\pi : K \rightarrow \text{Inn}K$ for the quotient and reserve $s : \text{Inn}K \rightarrow K$ for a chosen *set-section* with $s(e) = e$ (not assumed to be a homomorphism).

2.3 Gerbes, bands, and rectification

We use the gerbe vocabulary only to *name* the object and import two classical facts; the cocycles and all computations are derived in place.

A *gerbe* over $N(\mathcal{C})$ (for us: a K -gerbe) is a stack of groupoids that is locally nonempty and locally connected — locally equivalent to the trivial gerbe BK . Its local objects are *inventory trivialisations*; its *band* (lien) is the sheaf of automorphisms of those local objects. A gerbe has a *global section* (a global object) precisely when it is equivalent to the trivial gerbe. For an abelian band A , isomorphism classes of A -gerbes over $N(\mathcal{C})$ are classified by $H^2(N(\mathcal{C}); A)$, with the trivial class corresponding to the gerbe that admits a global section (Giraud [6], III; see also Breen [7] for the bitorsor/cocycle presentation). When the band is nonabelian the classification is Giraud’s full nonabelian H^2 ; we will not need it, because our scope hypothesis trivialises the band to its centre (Lemma 7).

A *strict global inventory* is a global section realised *on the nose*: a strict functor $\mathcal{C} \rightarrow BK$ whose transition data satisfy the gluing equations strictly rather than up to coherent isomorphism. The passage from a pseudofunctor (coherent, up-to-iso gluing) to a strict functor is *rectification*; that a pseudofunctor into a bicategory can be replaced by an equivalent strict one, and the obstruction to doing so compatibly with prescribed data, is the Bénabou–Street rectification circle of ideas [8, 9]. We cite these only to place the object; the obstruction class and the rectification algebra are computed in Sections 3–4.

3 The inventory gerbe: setup, band, and cocycle

3.1 The tier-two datum

Definition 5 (Tier-two inventory datum). Fix an inventory fibre F with $K = \text{Aut}(F)$. Suppose local inventory charts at the nodes of $\mathcal{C}$ are comparable only up to relabelling, i.e. the transition along an operation f is a K -torsor of chart-identifications rather than a single one. The gluing data is then a K -gerbe over $N(\mathcal{C})$: a stack whose local objects are inventory trivialisations and whose automorphism band is (a subsheaf of) K . A *strict global inventory* is a strict global section of this gerbe.

Definition 6 (Scope hypothesis: trivial band). We fix the *same* fibre F and relabelling group K at every node and *trivialise the outer action*: the band/lien of the gerbe is the trivial (constant) one. A nontrivial outer band twists the coefficients to a local system Z_ρ and introduces a prior Giraud band obstruction; that genuinely nonabelian case is outside this note and is marked open in Section 8.

Under Definition 6 the gerbe is *band-trivialised*, and the next lemma identifies the coefficients.

3.2 The band is the centre

Lemma 7 (Band = centre). *The band of a band-trivialised K -gerbe is the constant abelian sheaf $Z(K)$, with trivial $\mathcal{C}$ -action.*

Proof. The band is the sheaf of automorphisms of the gerbe's local objects. Two local trivialisations of the same fibre differ by an element of K , and an automorphism of a local object is realised by an element $z \in K$ acting on every chart; for the action to be independent of the choice of trivialisation — i.e. to descend to the band of the band-trivialised gerbe — z must commute with all of K , so $z \in Z(K)$. Equivalently, $Z(K)$ is the group of 2-automorphisms of the identity autoequivalence of the fibre (the Eckmann–Hilton centre of the automorphism 2-group of F [10, 11]). Inner automorphisms, which *can* be undone by re-choosing local trivialisations, fix $Z(K)$ pointwise, so the $\mathcal{C}$ -action on the band is trivial. This is the standard gerbe-band formalism (Giraud [6], III; Breen [7]); we import it rather than re-derive it. $\square$

So the coefficients are $Z(K) = Z(\text{Aut}F)$ with trivial action, and the gerbe is classified by $H^2(N(\mathcal{C}); Z(\text{Aut}F))$.

3.3 The cocycle in each face

There are two faces, according to whether $K = \text{Aut}(F)$ is abelian. Both have band $Z(K)$; both are needed, because the minimal witness is abelian while the smallest nonabelian witnesses are D_4 and Q_8 .

Face 1 — abelian relabelling (K abelian, $Z(K) = K$). Here the fibre is a K -torsor-stack: stock is defined up to the gauge group K , and the torsor itself only locally. The gerbe is a genuine abelian K -gerbe, presented by a normalised 2-cocycle $\omega \in Z^2(N(\mathcal{C}); K)$ (its transition cocycle), and $[\mathfrak{g}] := [\omega] \in H^2(N(\mathcal{C}); K)$. That a global section exists iff $[\omega] = 0$ is the defining property of the abelian-gerbe classification $H^2(N(\mathcal{C}); K) \cong \{K\text{-gerbes}\}/\simeq$, with the trivial gerbe the unique one admitting a section (Giraud [6]). The minimal $B(\mathbb{Z}/2)$ witness of Section 5 lives in this face.

Face 2 — nonabelian relabelling, trivial band (the lifting gerbe). When K is nonabelian the inner part of a relabelling composes strictly: re-trivialising a fibre conjugates the transport, so comparison of charts is canonical only up to inner automorphism, and the unambiguous datum is a genuine functor

$$\rho : \mathcal{C} \longrightarrow B(\text{Inn}K).$$

The gerbe is the *gerbe of lifts of ρ through π* in $(\star)$. Choose set-section lifts $T_f := s(\rho_f) \in K$ for each operation f . Because both $T_g T_f$ and T_{gf} map to $\rho_g \rho_f = \rho_{gf}$ under π , their ratio lands in $\ker \pi = Z$:

$$T_g T_f = \omega(g, f) \cdot T_{gf}, \quad \omega(g, f) \in Z. \quad (\dagger)$$

Thus $\omega = \rho^*(\text{extension 2-cocycle of } (\star))$ is a normalised 2-cochain. Associativity of multiplication in K , together with the centrality of Z (so every conjugation acts trivially on the values of ω), gives the normalised 2-cocycle identity

$$\omega(h, g) \omega(hg, f) = \omega(g, f) \omega(h, gf), \quad \text{i.e. } \delta\omega = 0,$$

whence $[\mathfrak{g}] := \rho^*[\star] = [\omega] \in H^2(N(\mathcal{C}); Z)$. Changing the section s replaces ω by a cohomologous cochain, so $[\mathfrak{g}]$ is well-defined.

In both faces $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(\text{Aut}F))$ and the rectification freedom is a 1-cochain valued in $Z(\text{Aut}F)$; the next section treats both uniformly.

4 Rectification is the vanishing of $[\mathfrak{g}]$

Theorem 8 (Theorem A). *A strict global inventory — a strict functor $\mathcal{C} \rightarrow BK$ realising the gluing of Definition 5 — exists if and only if $[\mathfrak{g}] = 0$ in $H^2(N(\mathcal{C}); Z)$, where $Z = Z(\text{Aut}F)$.*

Proof. Re-choosing the local trivialisation at each operation f multiplies the transition datum T_f by an element c_f of the band $Z = Z(\text{Aut}F)$. (In Face 2, $c_f \in Z$ moves the lift T_f within its Z -torsor of lifts of ρ_f ; in Face 1, with K abelian, $c_f \in K = Z$ rescales the abelian-gerbe transition directly.) A candidate strict inventory is therefore a 1-cochain $c : \text{Mor } \mathcal{C} \rightarrow Z$ producing new transitions $T'_f = c_f \cdot T_f$, and strictness is the demand that $T'_{gf} = T'_g T'_f$. Using $(\dagger)$ and the centrality of each c_f ,

$$c_{gf} T_{gf} = T'_{gf} \stackrel{!}{=} T'_g T'_f = c_g c_f T_g T_f = c_g c_f \omega(g, f) T_{gf},$$

which holds iff

$$\omega(g, f) = c_{gf} c_g^{-1} c_f^{-1} = (\delta c^{-1})(g, f).$$

Hence a strict global inventory exists iff ω is the coboundary of a Z -valued 1-cochain on $N(\mathcal{C})$, i.e. iff $[\mathfrak{g}] = [\omega] = 0$ in $H^2(N(\mathcal{C}); Z)$. The computation uses only that each c_f is central, so it is uniform across the two faces. $\square$

This is the central-extension lifting criterion (Face 2) and the abelian-gerbe triviality criterion (Face 1), both specialised to $N(\mathcal{C})$; Giraud and Breen are cited only to name the object, while the cocycle, its class, and the rectification algebra are derived above.

Remark 9 (Concrete check: D_4 and Q_8). Take $\mathcal{C} = B((\mathbb{Z}/2)^2)$ and $K \in \{D_4, Q_8\}$, each with $Z(K) = \mathbb{Z}/2$, $\text{Inn}K = (\mathbb{Z}/2)^2$, and $\rho = \text{id}$ (the inner transport is the full inner group). A strict lift would be a splitting homomorphism $(\mathbb{Z}/2)^2 \rightarrow K$ of $(\star)$; none exists, since K is nonabelian while $(\mathbb{Z}/2)^2$ is abelian. Correspondingly the section-cocycle ω of $(\dagger)$ is not a coboundary, so $[\mathfrak{g}] \neq 0$ and rectification fails — exactly when the class is nonzero. Both the non-splitting and $[\mathfrak{g}] \neq 0$ are confirmed by brute-force enumeration (`inventory_gerbe_h2.py`).

5 $[\mathfrak{g}]$ is genuinely a degree-two datum

Theorem 10 (Theorem B). (i) *If $N(\mathcal{C})$ is homotopy 1-truncated — in particular if $\mathcal{C}$ is the free category on a directed acyclic quiver (a tree-like or DAG supply chain) — then $H^2(N(\mathcal{C}); Z) = 0$, so $[\mathfrak{g}] = 0$ for every fibre F .* (ii) *There is a minimal one-node witness, $\mathcal{C} = B(\mathbb{Z}/2)$ with $K = \mathbb{Z}/2$, for which $[\mathfrak{g}] \neq 0$.*

Proof of (i). The classifying space of the free category $F(Q)$ on a quiver Q is homotopy equivalent to $|Q|$, a 1-complex (no relations means no higher cells up to homotopy), so $H^{\geq 2}(N(F(Q)); \mathbb{Z}) = 0$. It is worth seeing this directly on the reconvergent *diamond* $a \rightrightarrows \{b, c\} \rightrightarrows d$ — the paradigmatic two-route chain — because its nerve is *not* literally 1-dimensional: it carries two nondegenerate 2-simplices, the composites $p; r$ and $q; s$ of the two routes. A direct simplicial computation over $\mathbb{F}_2$ (`diamond_nerve_h2.py`) gives chain dimensions $(C^0, C^1, C^2) = (4, 6, 2)$, boundary ranks $\text{rk } d^0 = 3$ and $\text{rk } d^1 = 2$, and hence

$$H^0 = 1, \quad H^1 = 1, \quad H^2 = 0, \quad \chi = 4 - 6 + 2 = 0.$$

The lone H^1 is precisely the tier-one reconvergence loop $[\theta_R]$; $H^2 = 0$ says there is *no gerbe on a DAG*, even one with genuine 2-simplices. $\square$

Proof of (ii). For $G = \mathbb{Z}/2$ and coefficients $\mathbb{Z}/2$ one has $H^n(\mathbb{Z}/2; \mathbb{Z}/2) = \mathbb{Z}/2$ for all $n \geq 0$ (`inventory_gerbe_h2.py`); in particular $H^2(\mathbb{Z}/2; \mathbb{Z}/2) = \mathbb{Z}/2 \neq 0$. The nonzero class is the extension $1 \rightarrow \mathbb{Z}/2 \rightarrow \mathbb{Z}/4 \rightarrow \mathbb{Z}/2 \rightarrow 1$. Concretely, a reversible two-lot swap g (with $g^2 = \text{id}$ in $\mathcal{C}$) has a relabelling transport T_g that cannot be chosen to satisfy $T_g^2 = \text{id}$ strictly: rescaling T_g by $a \in \mathbb{Z}/2$ changes T_g^2 by $2a = 0$, so the “square-root ambiguity” cannot be absorbed — one would need $\mathbb{Z}/4$ to lift. Hence $[\mathbf{g}] \neq 0$. $\square$

Thus $[\mathbf{g}]$ lives at cohomological degree 2 (nerve level N_3), vanishing on every acyclic chain yet already nonzero on a single reversible two-lot loop. For the record, the richer witness $H^2((\mathbb{Z}/2)^2; \mathbb{Z}/2) = \mathbb{F}_2^3$ shows that $(\mathbb{Z}/2)^2$ carries three independent $\mathbb{Z}/2$ -gerbes — the D_4 , Q_8 , and $\mathbb{Z}/4 \times \mathbb{Z}/2$ classes — of which the first two were used in Remark 9.

6 Independence from the Zappa–Szép weld

Theorem 11 (Theorem C). $[\mathbf{g}] \in H^2(N(\mathcal{C}); Z(\text{Aut} F))$ and the Zappa–Szép weld $[\omega] \in H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D})$ are logically independent: neither vanishing implies the other. They are classes of the same degree over different coefficient systems — the inventory-fibre centre $Z(\text{Aut} F)$ versus the weld vertex-group presheaf $\mathcal{D}$ — so, unlike the tier-one H^1 versus H^2 separation, the distinction is not by degree but by coefficients, and must be checked by witnesses.

Proof. The weld $[\omega]$ is determined by $\mathcal{C}$ alone: it obstructs the closing of a Zappa–Szép transversal, i.e. whether $\mathcal{C}$ factors as procurement $\bowtie$ logistics [3, 4]. The gerbe $[\mathbf{g}]$ depends on the inventory fibre F , which is data *external* to $\mathcal{C}$. For any fixed $\mathcal{C}$: choosing $F = \text{pt}$ (so $\text{Aut} F = 1$, strict inventory) forces $[\mathbf{g}] = 0$ whatever $[\omega]$ is; choosing $\mathcal{C} = B(\mathbb{Z}/2)$ with $\text{Aut} F = \mathbb{Z}/2$ forces $[\mathbf{g}] \neq 0$ (Theorem 10) regardless of the weld. Since the two classes obstruct different data, no implication holds. We make both directions explicit.

Witness W'_1 ($[\mathbf{g}] \neq 0$, $[\omega] = 0$). Let $\mathcal{C} = B(\mathbb{Z}/2)$, one reversible operation, with the trivial Zappa–Szép factorisation $\mathcal{D} = 0$, forcing $[\omega] \in H^2(\text{Sk}_{\mathcal{C}}; 0) = 0$. Take inventory relabelling $K = \mathbb{Z}/2$; then $[\mathbf{g}]$ is the generator of $H^2(\mathbb{Z}/2; \mathbb{Z}/2) = \mathbb{Z}/2 \neq 0$ (Theorem 10). The chain carries a non-rectifiable relabelling gerbe but no weld structure at all.

Witness W'_2 ($[\mathbf{g}] = 0$, $[\omega] \neq 0$). Let $\text{Sk}_{\mathcal{C}} = (a \rightarrow x \rightrightarrows y)$ be the rigid-twist skeleton with $\mathcal{D} = \mathbb{Z}/2$ and $[\omega]$ the generator of $H^2(\text{Sk}_{\mathcal{C}}; \mathbb{Z}/2) = \mathbb{Z}/2 \neq 0$ [3], equipped with strict inventory ($F = \text{pt}$, $\text{Aut} F = 1$), so $[\mathbf{g}] = 0$. The chain fails to factor, yet its (trivial-relabelling) inventory rectifies vacuously. $\square$

The coefficient systems genuinely differ: in W'_1 the weld coefficient $\mathcal{D} = 0$ while the gerbe coefficient $Z(\text{Aut}F) = \mathbb{Z}/2 \neq 0$. There is no natural map comparing $H^2(N(\mathcal{C}); Z(\text{Aut}F))$ with $H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D})$, let alone an identification — the same “rhyme, not identity” trap as for the two distinct $[\omega]$ -sites studied earlier [5], but now with *matching* degree, so the coefficient system is the only discriminator and must be examined.

Remark 12 (Relation to tier one). The tier-one class $[\theta_R] \in H^1$ and the tier-two class $[\mathfrak{g}] \in H^2$ differ in degree, hence are immediately independent. The full inventory obstruction is the two-tier filtration with associated graded $H^1(N(\mathcal{C}); R)$ (counts) and $H^2(N(\mathcal{C}); Z(\text{Aut}F))$ (relabellings), with $[\omega] \in H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D})$ a separate structural precondition. Whether the two tiers combine as a split sum or a nontrivial extension (a d_3 -type coupling $H^1 \rightarrow H^3$) is not determined here; see Section 8.

7 When the gerbe is nonvacuous

A gerbe needs nonzero centre. This is a genuine — and slightly surprising — constraint on *which* fibres carry a tier-two obstruction.

Proposition 13 (Proposition D). $[\mathfrak{g}]$ can be nonzero only if $Z(\text{Aut}F) \neq 1$. In particular, for F a bare n -element set of indistinguishable lots, $\text{Aut}(F) = S_n$ with

$$Z(S_n) = 1 \ (n \geq 3), \quad Z(S_2) = \mathbb{Z}/2, \quad Z(S_n) = 1 \ (n \leq 1),$$

so the tier-two gerbe is *vacuous* (always $[\mathfrak{g}] = 0$) for $n \geq 3$ fungible lots and *nonvacuous* only for $n = 2$.

Proof. By Lemma 7 the coefficients are $Z(\text{Aut}F)$, and $H^2(N(\mathcal{C}); 1) = 0$, so $[\mathfrak{g}] = 0$ whenever $Z(\text{Aut}F) = 1$. The centre table for S_n ($n \leq 5$) is confirmed by direct computation (`inventory_gerbe_h2.py`); $Z(S_n) = 1$ for $n \geq 3$ is classical. $\square$

So a nontrivial relabelling gerbe requires either *exactly two* fungible lots, or a *structured* fibre whose automorphism group has a centre: e.g. F a K -torsor (stock defined up to a gauge group K , automorphisms acting by translation, centre $Z(K)$ — this is exactly the torsor-valued inventory the companion note anticipated), or graded/typed lots with central symmetries (e.g. $\text{Aut}F = \text{GL}_k$, centre the scalars). The interpretation is economic: generic fungibility of ≥ 3 interchangeable units creates *no* degree-two obstruction — any relabelling ambiguity is absorbed by inner reparametrisation, because S_n is centreless — while two-fold or gauge/torsor-structured ambiguity does. The tier-two case that matters for applications is therefore gauge-valued stock (allocation up to a group action), not plain fungibility.

8 Scope, open problem, and the two-tier bottom line

Refining inventory comparison from *equality of counts* to *isomorphism of stock-states* raises the consistency obstruction by exactly one categorical level: from the degree-one count-discrepancy class $[\theta_R] \in H^1(N(\mathcal{C}); R)$ to a degree-two gerbe class $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(\text{Aut}F))$. A strict global inventory exists iff $[\mathfrak{g}] = 0$ (Theorem 8); $[\mathfrak{g}]$ is genuinely degree-two, vanishing on every DAG or tree chain but nonzero already on a single reversible two-lot loop $B(\mathbb{Z}/2)$ (Theorem 10); it is independent of the Zappa–Szép weld $[\omega]$, with which it shares a degree but not a coefficient system (Theorem 11); and it is nonvacuous exactly when $Z(\text{Aut}F) \neq 1$ — two fungible lots or gauge/torsor-structured stock, not generic ≥ 3 -fold fungibility (Proposition 13). Inventory consistency is thus a two-tier

theory: H^1 **counts**, H^2 **relabellings**, over a structural precondition $[\omega]$. Within the container program this is the fourth distinct H^2 home on $\mathbf{DCont} \simeq \mathbf{Cat}$, after Zappa–Szép holonomy, Ferri prunability, and the Baues–Wirsching class — the first to be distinguished by its coefficient system rather than its site.

Scope hypotheses (stated honestly). Two hypotheses bound the result.

1. *Trivial band* (Definition 6): the same fibre F and relabelling group K at every node, with the outer action trivialised. The computations above all live under this hypothesis.
2. $[\mathbf{g}]$ is the *rectification* obstruction — the existence of a strict global section. Classifying all tier-two inventories up to equivalence (not merely their existence) is Giraud’s full nonabelian H^2 , a torsor over $H^2(N(\mathcal{C}); Z)$ once nonempty; that is not treated here.

The open problem (not claimed). A *nontrivial outer band* — different fibres or a nontrivial $\mathcal{C}$ -action on K — twists the coefficients to a local system Z_ρ and introduces a *prior* Giraud band obstruction, a genuinely nonabelian H^2 class whose vanishing is necessary before the abelian rectification class $[\mathbf{g}]$ is even defined. We do not have this band obstruction; determining it for directed-container supply chains with heterogeneous fibres is left open. Relatedly (see Remark 12), the coupling between tiers one and two — whether they split or form a Postnikov-type extension via a d_3 map $H^1 \rightarrow H^3$ — is stated as a filtration but not computed; a bar-complex oracle for the composite nerve $B(\mathbb{Z}/2)$ glued to the diamond is the natural next verification.

For the grant. The worked economic example to develop is the torsor-fibre case $\mathrm{Aut} F = K$ acting on itself, with $Z(\mathrm{Aut} F) = Z(K)$ — gauge-valued allocation, where stock is defined only up to a group action and the tier-two obstruction has real content. This is the applications payoff of Proposition 13: the economically relevant non-rectifiable inventory is not the many-interchangeable-units case but the gauge-structured one.

References

- [1] D. Ahman, T. Uustalu, *Directed containers as categories*, Proc. MSFP 2016, EPTCS 207, 2016, 89–98; arXiv:1604.01187.
- [2] MacBeth, *Supply-chain inventory consistency is a degree-one class, and is independent of the Zappa–Szép weld obstruction* (Kodamai internal note, 6 October 2026; proved).
- [3] MacBeth, *The global-closure obstruction (G) is a cohomology class* (Kodamai internal note, 11 June 2026; proved, Lean-adjacent).
- [4] MacBeth, *The re-entrancy obstruction is the token-mutation bit: an analytic proof for orchestration Zappa–Szép products* (Kodamai internal note, 20 July 2026; proved, $[\omega] = \varepsilon$ core Lean-verified).
- [5] MacBeth, *Two $[\omega]$ -sites are irreducibly distinct* (Kodamai internal note, 14 August 2026; proved).
- [6] J. Giraud, *Cohomologie non abélienne*, Grundlehren der mathematischen Wissenschaften **179**, Springer, 1971.

- [7] L. Breen, *Bitorseurs et cohomologie non abélienne*, in *The Grothendieck Festschrift, Vol. I*, Progr. Math. **86**, Birkhäuser, 1990, 401–476.
- [8] J. Bénabou, *Introduction to bicategories*, in *Reports of the Midwest Category Seminar*, Lecture Notes in Math. **47**, Springer, 1967, 1–77.
- [9] R. Street, *Categorical structures*, in *Handbook of Algebra, Vol. 1*, North-Holland, 1996, 529–577.
- [10] J. C. Baez, A. D. Lauda, *Higher-dimensional algebra V: 2-groups*, Theory Appl. Categ. **12** (2004), 423–491.
- [11] N. Johnson, D. Yau, *2-Dimensional Categories*, Oxford University Press, 2021.
- [12] H.-J. Baues, G. Wirsching, *Cohomology of small categories*, J. Pure Appl. Algebra **38** (1985), 187–211.
- [13] C. A. Weibel, *An Introduction to Homological Algebra*, Cambridge Studies in Advanced Mathematics **38**, Cambridge Univ. Press, 1994.